Capstone Project Submission

Line Following Robot with PID Control

Course: Digital Twin Course

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1 Introduction

This technical report presents the modeling, control, and experimental validation of a 2-wheel differential-drive mobile robot digital twin performing path tracking on a two-dimensional plane, which is implemented using the Innexis Virtual System Interconnect (VSI) backplane.

2 System Architecture

2.1 VSI Communication Configuration

The distributed digital twin was implemented using the **Innexis Virtual System Interconnect (VSI)** backplane. Communication between components was realized using the `python2DtEthernet` gateway over TCP/IP sockets.

Clients

Three clients were defined as a Python scripts implemented inside the user custom code region in the generated code of the digital twin system:

- **simulator** – plant and environment model
- **controller** – PID controller algorithm
- **visualizer** – real-time monitoring and KPI logging

Network Configuration

Each component was assigned a unique MAC and IP address:

Component	MAC Address	IP Address
Simulator	12:34:56:78:9A:01	192.168.1.1
Controller	12:34:56:78:9A:02	192.168.1.2
Visualizer	12:34:56:78:9A:03	192.168.1.3

Table 1: Virtual Ethernet Configuration

Socket Configuration

Two TCP/IP server-client connections were created:

- Port 8070: Simulator (Server) -¿ Controller (Client)
- Port 8071: Simulator (Server) -¿ Visualizer (Client)

This configuration enables:

- Closed-loop control communication between simulator and controller
- Passive signal tapping by the visualizer

Signal Definition

The simulator exposes the following signals:

- Outputs: $x, y, \theta, ref_x, ref_y$
- Inputs: vR_cmd, vL_cmd

The controller receives pose and reference signals and outputs wheel commands:

- Inputs: $x, y, \theta, ref_x, ref_y$
- Outputs: vR_cmd, vL_cmd

The visualizer subscribes to pose and reference signals only.

Signal Routing

Signal routing was configured as follows:

- Simulator pose -¿ Controller
- Simulator reference -¿ Controller
- Controller wheel commands -¿ Simulator
- Simulator pose and reference -¿ Visualizer

This structure forms a closed-loop feedback system with external monitoring.

Simulation Parameters

The global digital twin configuration was:

Digital Twin Name	`lineFollowerTwin`
Workspace Directory	diffDriveLineFollower/
Simulation Step	0.01 s
Total Simulation Time	100 s

Table 2: Global VSI Simulation Configuration

The simulation step of 10 ms ensures stable numerical integration while maintaining real-time communication consistency across the three distributed components.

2.2 Three-Client VSI Architecture

The system is composed of three clients and one server connecting them together for communication through Innexis VSI Simulator Control backplane:

- **Client 1 – Simulator (Plant + Environment)**

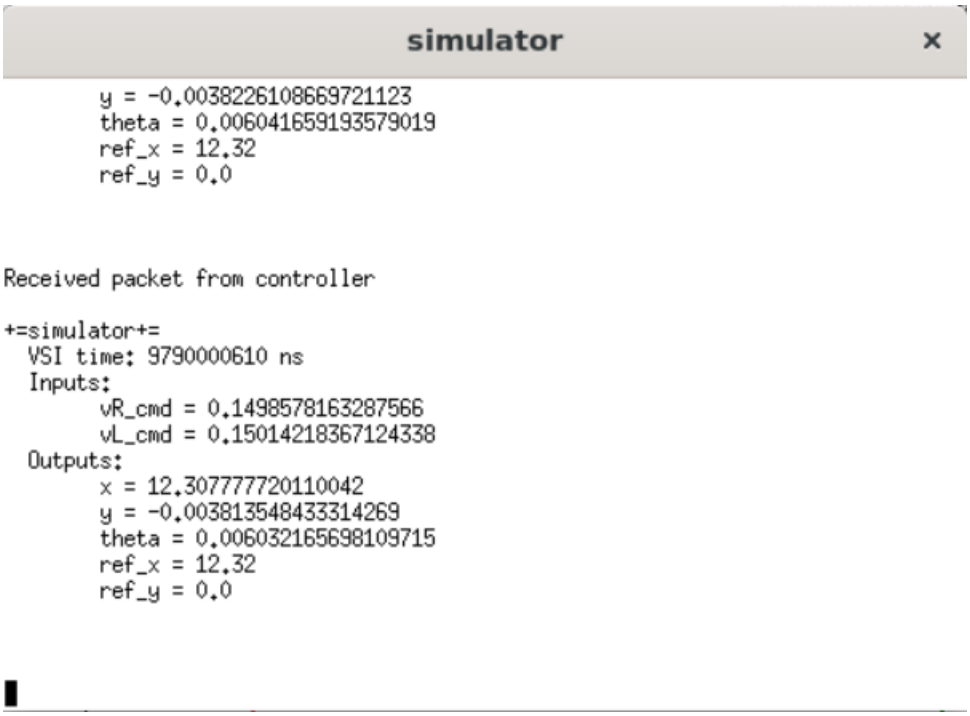

Figure 1: Simulator clinet widnow while running

- **Client 2 – Controller (PID)**

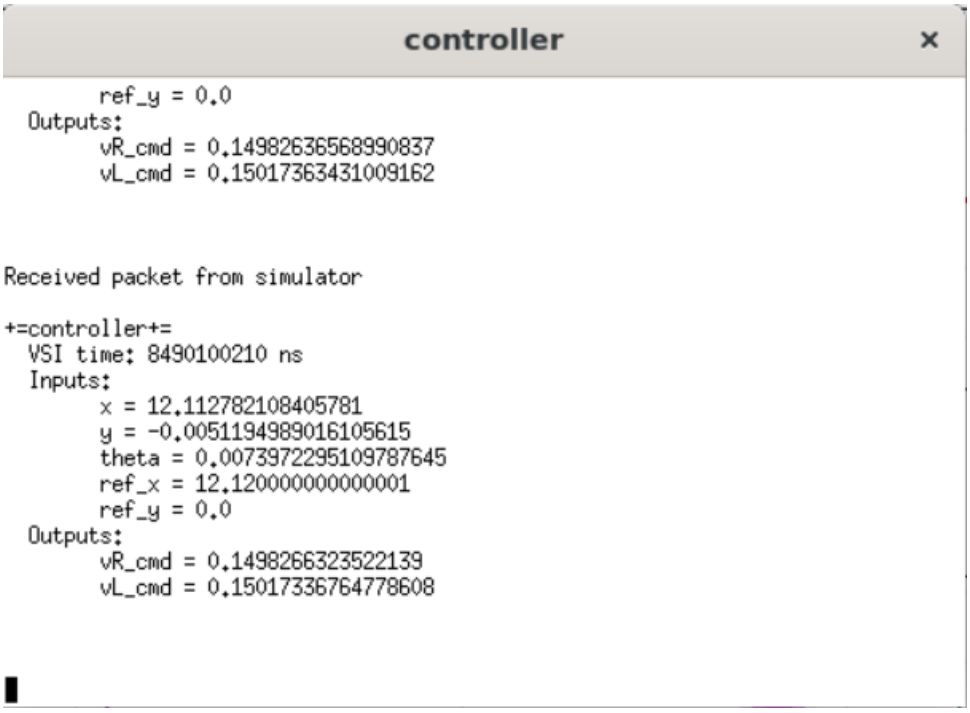

Figure 2: Controller clinet widnow while running

- **Client 3 – Visualizer/Logger**

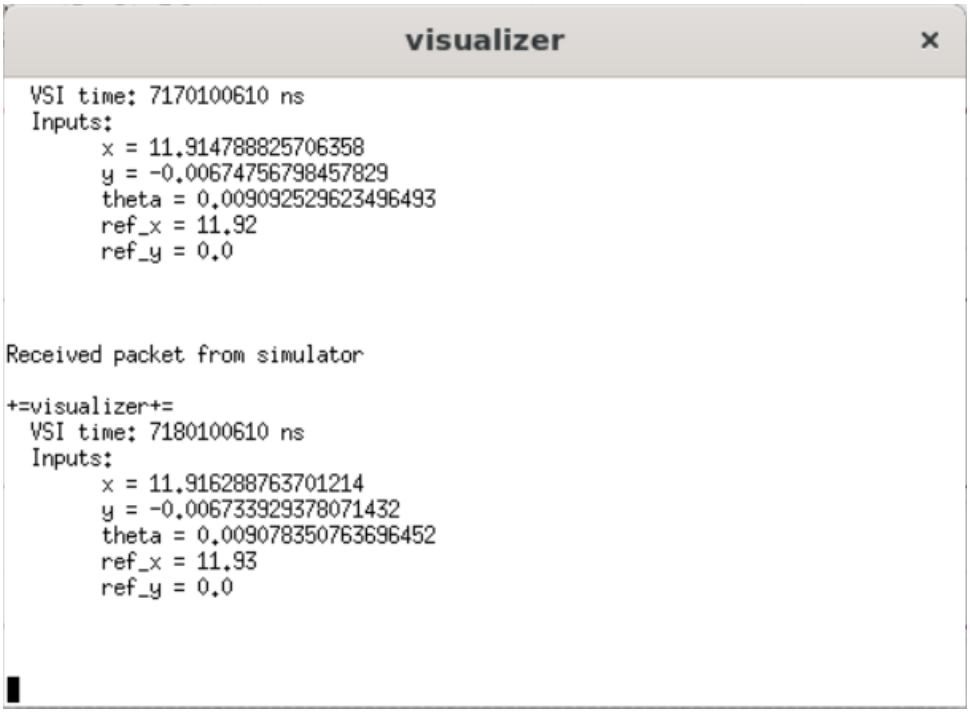

Figure 3: Visualizer clinet widnow while running

- **Server – Innexis VSI Simulator Control (Communication)**

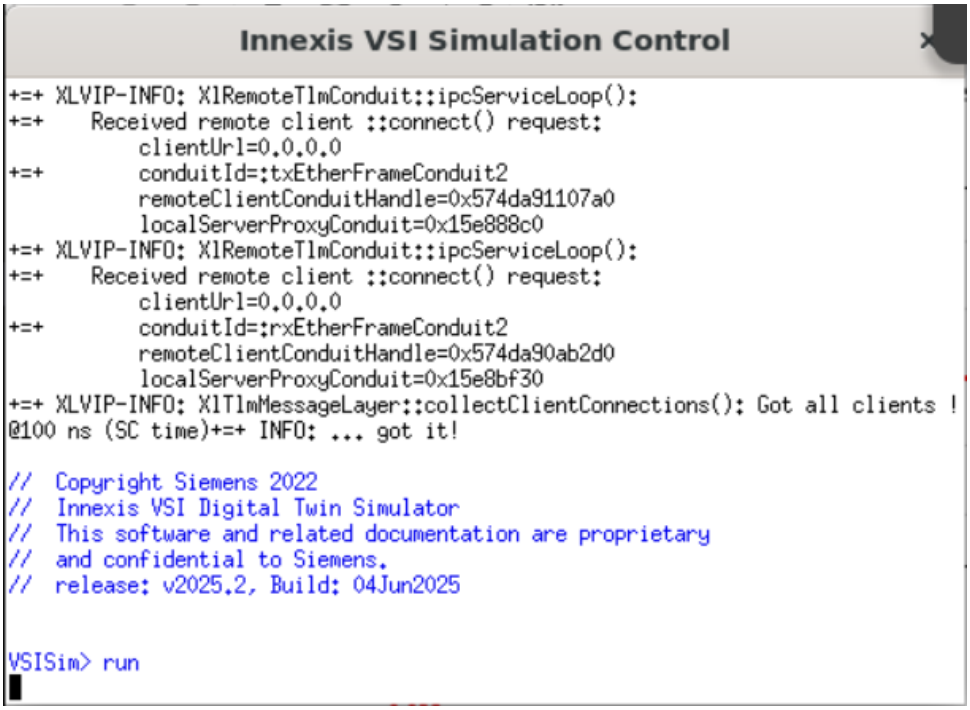

Figure 4: Innexis VSI Simulator Control Server widnow while running

3 Mathematical Modeling

3.1 Differential Drive Kinematics

The model assumes pure rolling without lateral slip and negligible wheel dynamics. Let:

$$v_R, v_L = \text{right and left wheel linear velocities}$$

$$L = \text{wheel base}$$

The robot linear and angular velocities are:

$$v = \frac{v_R + v_L}{2} \tag{1}$$

$$\omega = \frac{v_R - v_L}{L} \tag{2}$$

3.2 Continuous-Time Pose Model

$$\dot{x} = v\cos(\theta) \tag{3}$$

$$\dot{y} = v\sin(\theta) \tag{4}$$

$$\dot{\theta} = \omega \tag{5}$$

3.3 Discrete-Time Integration

Using Euler integration:

$$x_{k+1} = x_k + v\cos(\theta_k)\Delta t \tag{6}$$
$$y_{k+1} = y_k + v\sin(\theta_k)\Delta t \tag{7}$$
$$\theta_{k+1} = \theta_k + \omega\Delta t \tag{8}$$

Heading is wrapped to:

$$\theta \in (-\pi, \pi]$$

4 Lateral Error Formulation

Reference heading:

$$\theta_{ref} = \tan^{-1}\left(\frac{y_{ref} - y_{prev}}{x_{ref} - x_{prev}}\right) \tag{9}$$

Lateral error:

$$e_{lat} = (x - x_{ref})\sin(\theta_{ref}) - (y - y_{ref})\cos(\theta_{ref}) - (\theta - \theta_{ref}) \tag{10}$$

5 PID Controller Design

5.1 Control Law

$$\omega = K_P e_{lat} + K_I \int e_{lat} dt + K_D \frac{de_{lat}}{dt} \tag{11}$$

Inverse differential kinematics:

$$v_R = v + \omega\frac{L}{2} \tag{12}$$
$$v_L = v - \omega\frac{L}{2} \tag{13}$$

5.2 Discrete-Time PID Control Law

Since the controller is implemented in a discrete-time simulation with sampling period $T_s = 0.01$ s, the PID control law is formulated in discrete form.
Let the lateral error at time step k be:

$$e[k]$$

The discrete integral term is computed using accumulation:

$$I[k] = I[k-1] + e[k]T_s$$

The discrete derivative term is approximated using backward difference:

$$D[k] = \frac{e[k] - e[k-1]}{T_s}$$

The control signal at time step k is therefore:

$$u[k] = K_p e[k] + K_i I[k] + K_d D[k]$$

Substituting the discrete expressions:

$$u[k] = K_p e[k] + K_i \sum_{i=0}^{k} e[i] T_s + K_d \frac{e[k] - e[k-1]}{T_s}$$

For the PD controller used in Experiment E4, the integral gain is set to zero:

$$K_i = 0$$

which results in:

$$u[k] = K_p e[k] + K_d \frac{e[k] - e[k-1]}{T_s}$$

The control output $u[k]$ is then converted into differential wheel velocity commands for the left and right wheels.

Note: Wheel velocity saturation is applied to enforce actuator limits.

6 Simulation Configuration

Parameter	Value
Simulation Step	10 ms
Total Time	100 s
Wheel Base	0.3 m
Nominal Speed	1.5 m/s

Table 3: Common Simulation Parameters

7 Performance Evaluation Criteria

The controller performance was evaluated using the following key performance indicators (KPIs):

- Maximum overshoot (maximum absolute lateral deviation).
- Settling time (time required to remain within 2% of steady-state value, but as the required steady-state value is Zero, we change it to be 50% of the maximum value).
- Steady-state error (mean absolute error after convergence for last 20% of the errors).

Each experiment was conducted under identical simulation conditions to ensure fair comparison between gain sets.

8 Experiment E1 – PID Gain Sweep on Straight Reference Path

8.1 Gain Sets

Set	K_P	K_I	K_D
(1)	5.2	0.08	0.15
(2)	2.5	0.02	0.08
(3)	3.8	0.05	0.12
(4)	6.1	0.02	0.05
(5)	6.1	0.12	0.05

Table 4: PID Gain Sets

8.2 Trajectory and Error Plots on straight path reference

Parameter	Init$_V$$alue$
Initial X position	random(0.0, 20.0)
Initial Y position	0.0
Initial Heading Error	±60°
Nominal Speed	1.5 m/s

Table 5: Parameters Initialization

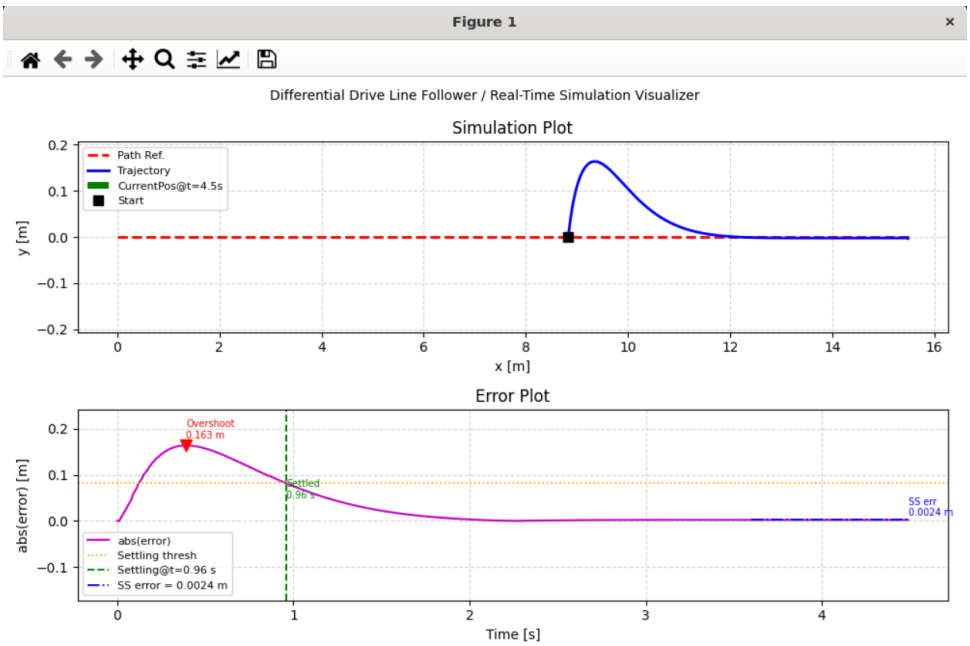

Figure 5: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (1), Trial 1

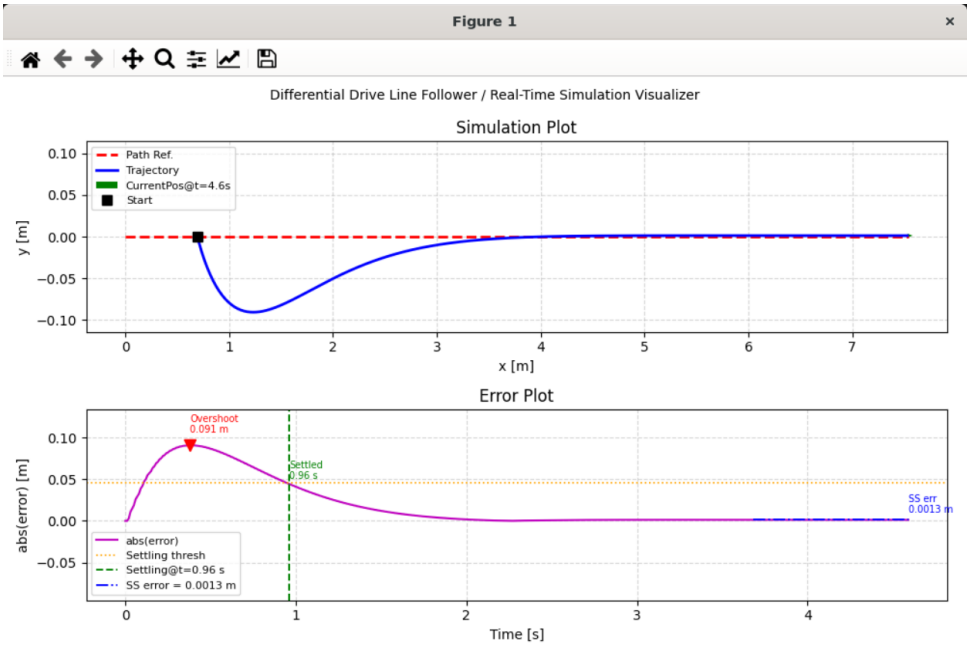

Figure 6: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (1), Trial 2

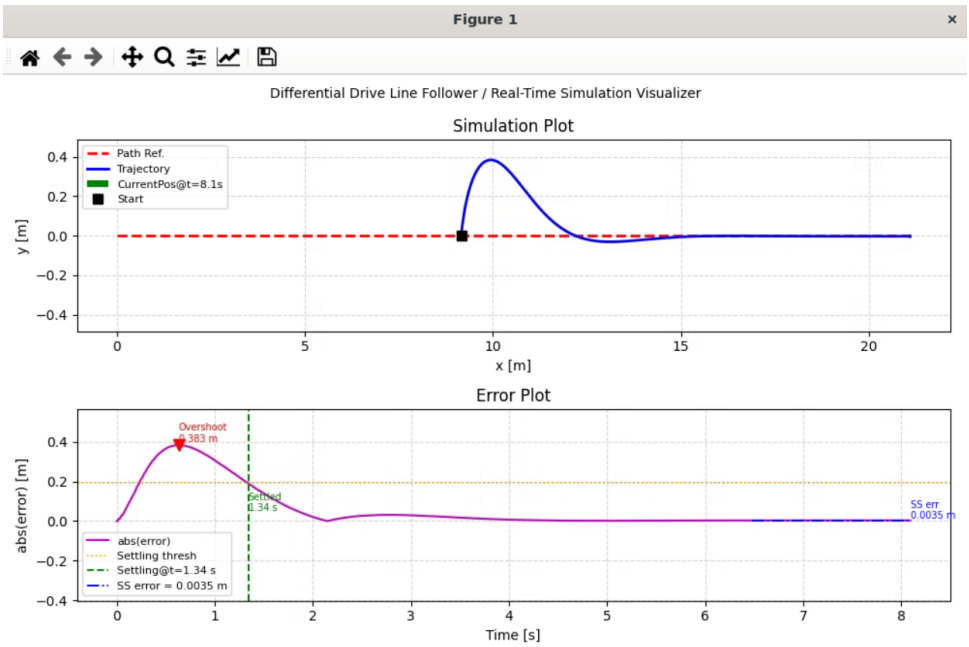

Figure 7: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (2), Trial 1

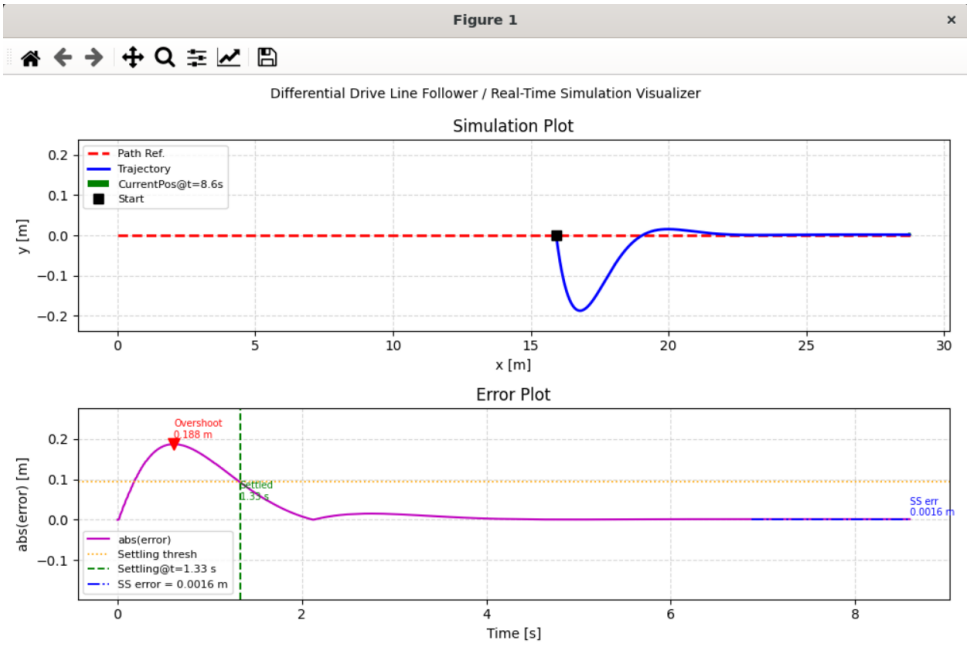

Figure 8: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (2), Trial 2

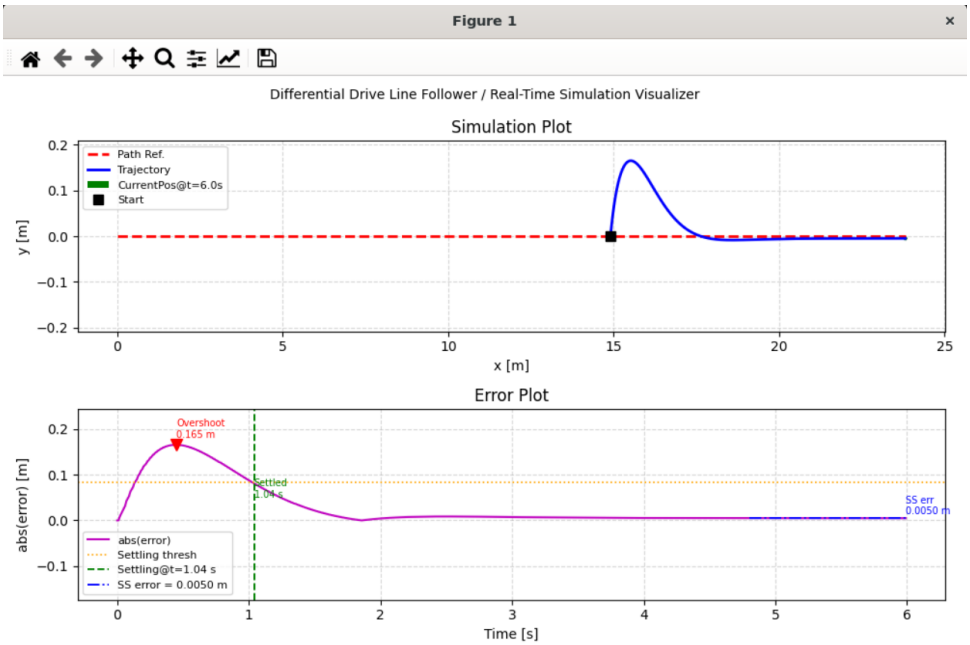

Figure 9: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (3), Trial 1

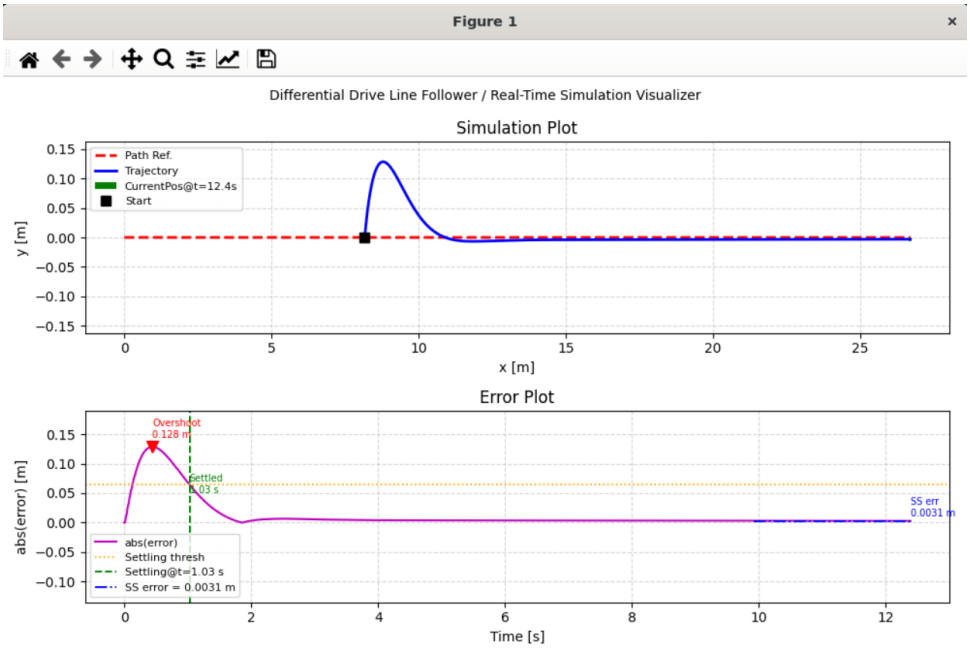

Figure 10: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (3), Trial 2

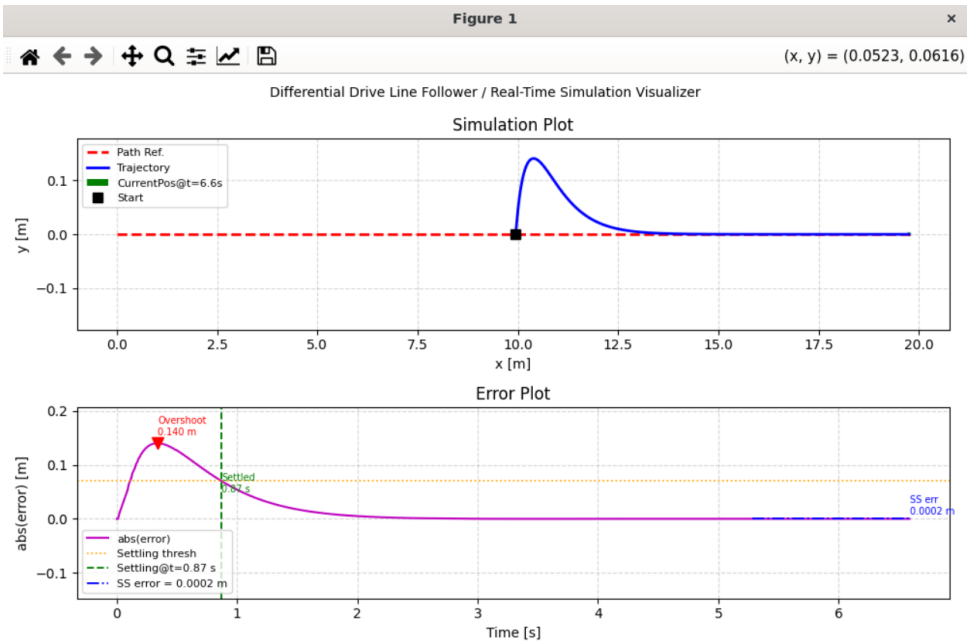

Figure 11: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (4), Trial 1

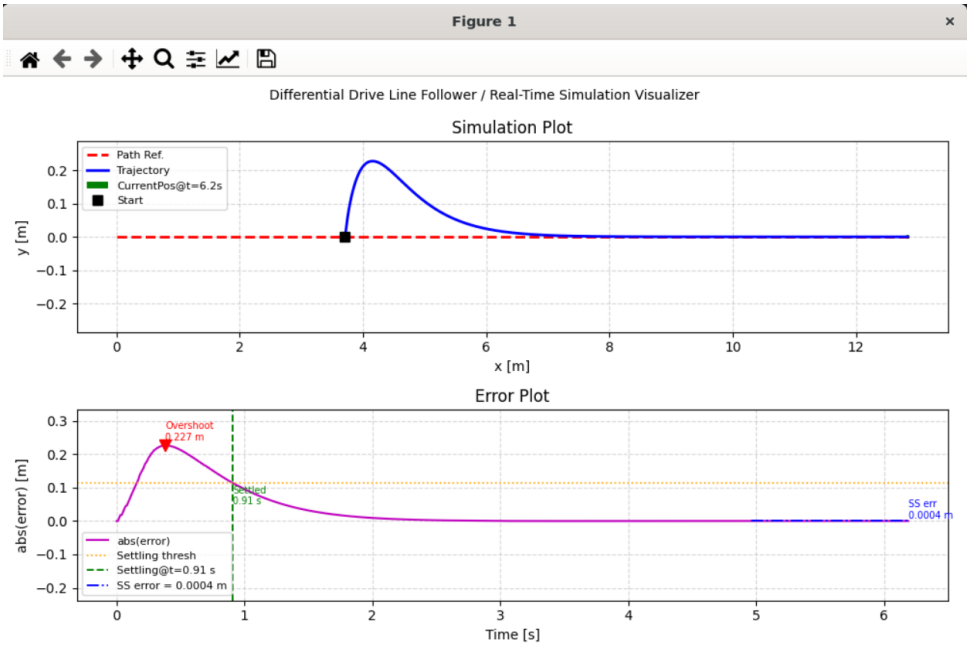

Figure 12: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (4), Trial 2

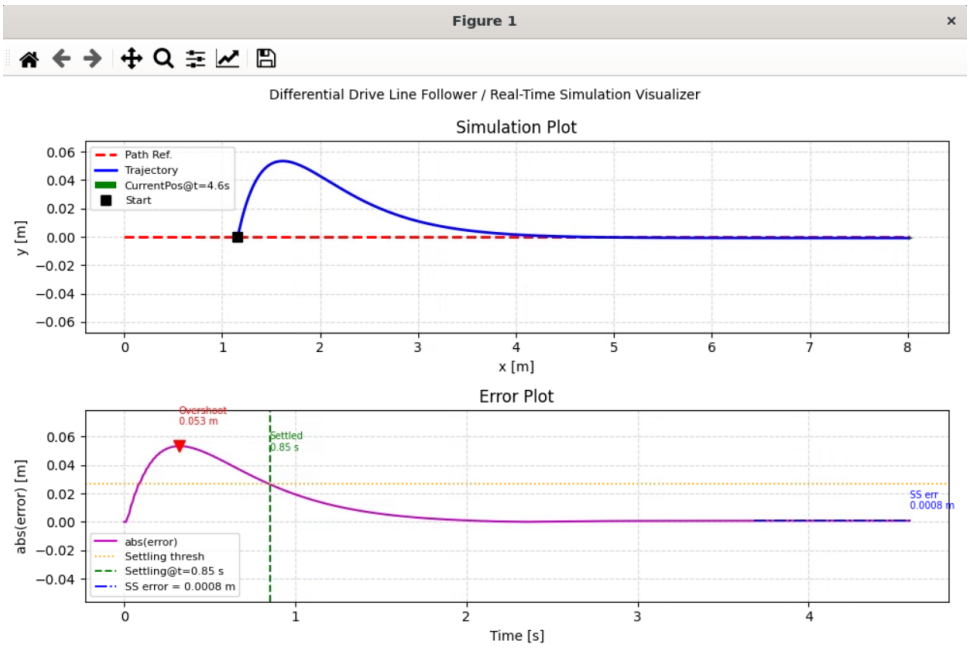

Figure 13: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (5), Trial 1

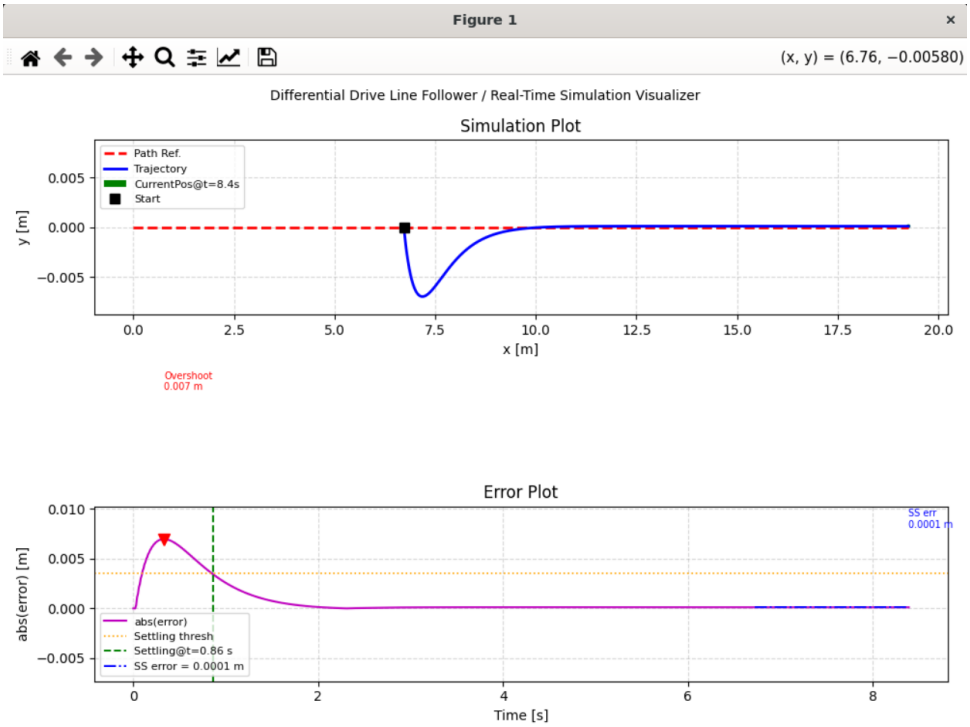

Figure 14: Trajectory Tracking Simulation // Lateral error Vs. Time – Gain Set (5), Trial 2

8.3 KPI Table

Gain Set	Overshoot (m)	Settling Time (s)	Steady-State Error (m)
(1)	0.127	0.96	0.00185
(2)	0.2855	1.335	0.00255
(3)	0.142	1.035	0.00405
(4)	0.1835	0.89	0.0003
(5)	0.03	0.855	0.00045

Table 6: E1. Performance Metrics

Discussion: Gain Set (5) achieved the lowest overshoot and fastest settling time, while Gain Set (4) showed the lowest steady-state error but with higher overshoot and slightly higher settling time. Though, Gain Set (5) significantly reduced overshoot due to the derivative term, demonstrating the benefit of the derivative term in reducing the overshoot and damping the system.

9 Experiment E2 – Curved Path Robustness

9.1 Curved Reference Path Generation

The curved reference trajectory was generated using a piecewise cubic Bézier curve chain forming an S-shaped profile.
A cubic Bézier curve is defined as:

$$\mathbf{B}(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3, \quad t \in [0,1] \tag{14}$$

where P_0, P_1, P_2, P_3 are control points.
The full reference path consists of five connected Bézier segments:

- Straight approach
- Right sweep
- Left S-turn
- Return to axis
- Straight exit

Each segment shares its end point with the subsequent segment to ensure positional continuity.
The resulting trajectory provides varying curvature regions, making it suitable for evaluating controller robustness under non-constant heading references.

9.2 Results & Analysis

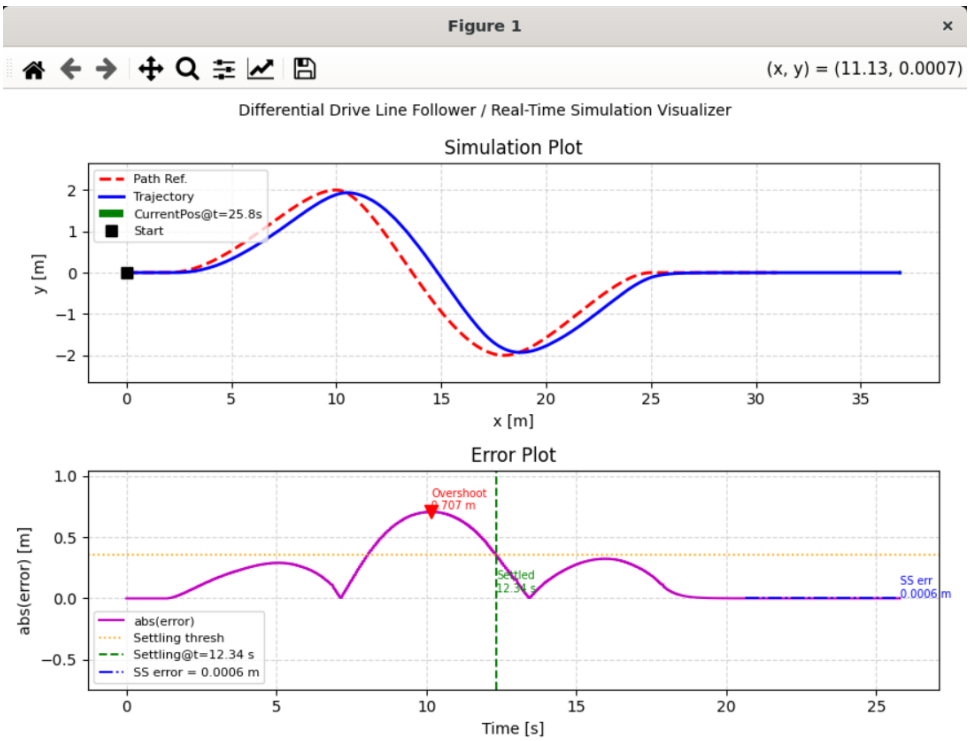

Figure 15: Curved Path Tracking

Curvature Effect Analysis: Compared to the straight-path experiment (E1), tracking error increased during high-curvature segments of the Bézier trajectory due to rapid heading variation. This highlights the controller sensitivity to curvature-induced lateral acceleration demands.

10 Experiment E3 – Noise and Disturbance Rejection

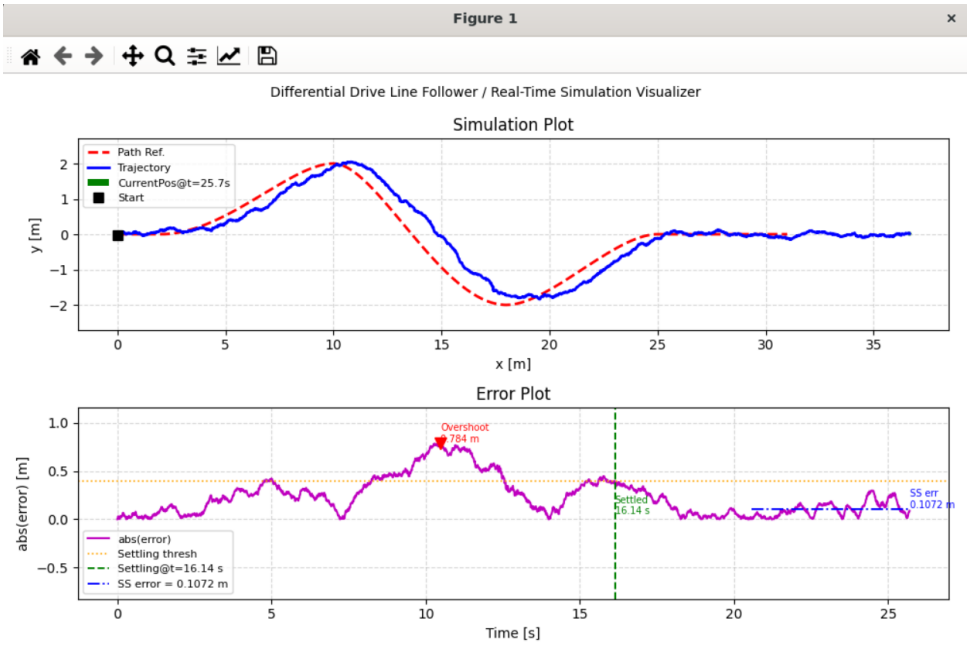

Figure 16: curved path tracking under low-level noise (standard deviation = 0.01m)

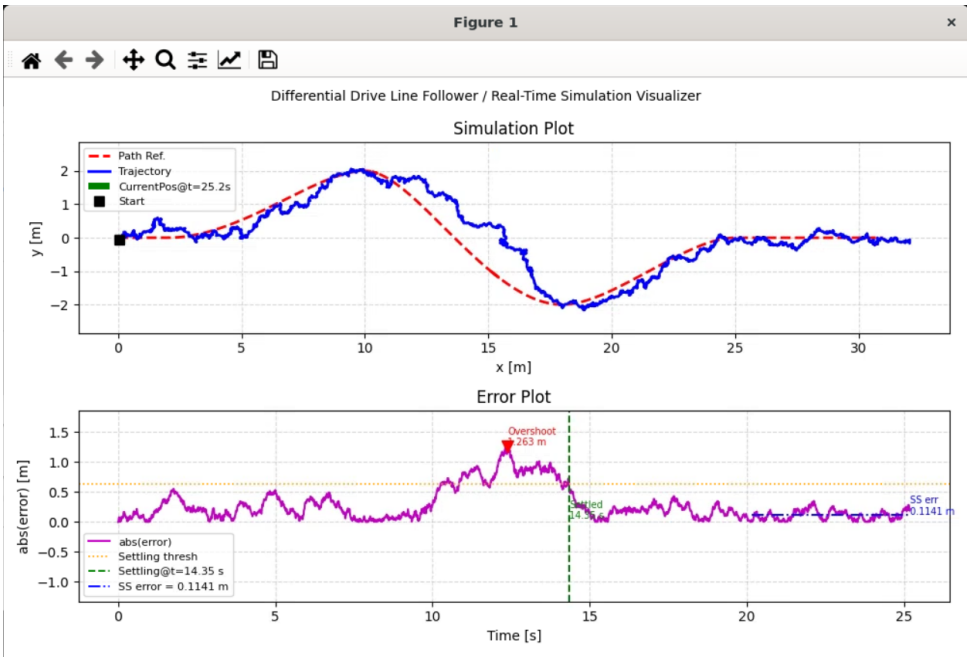

Figure 17: curved path tracking under moderate-level noise (standard deviation = 0.025m)

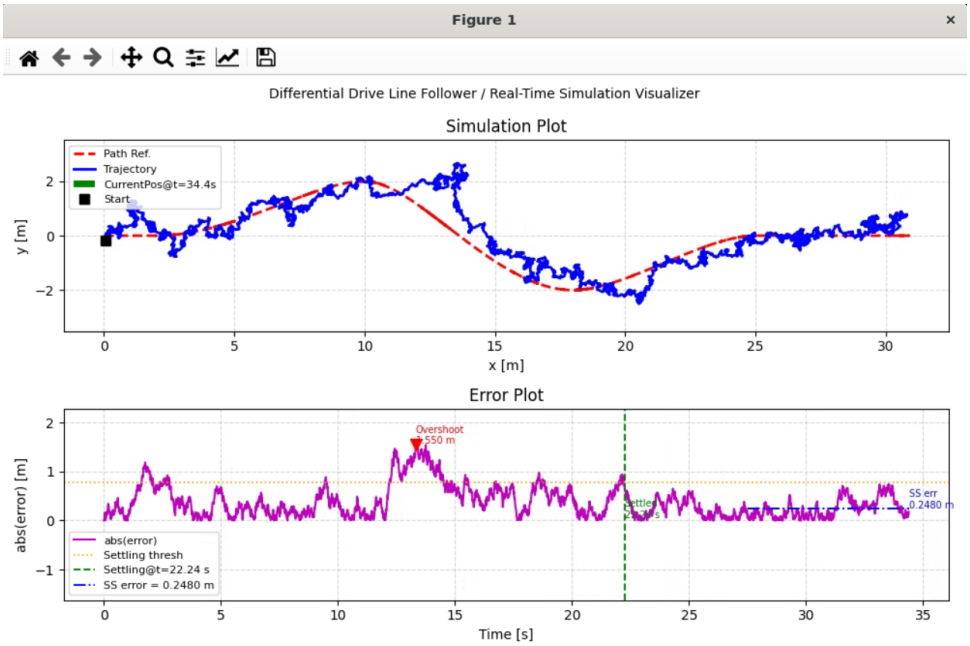

Figure 18: curved path tracking under high-level noise (standard deviation = 0.5m)

11 Experiment E4 – PD vs PID Comparison

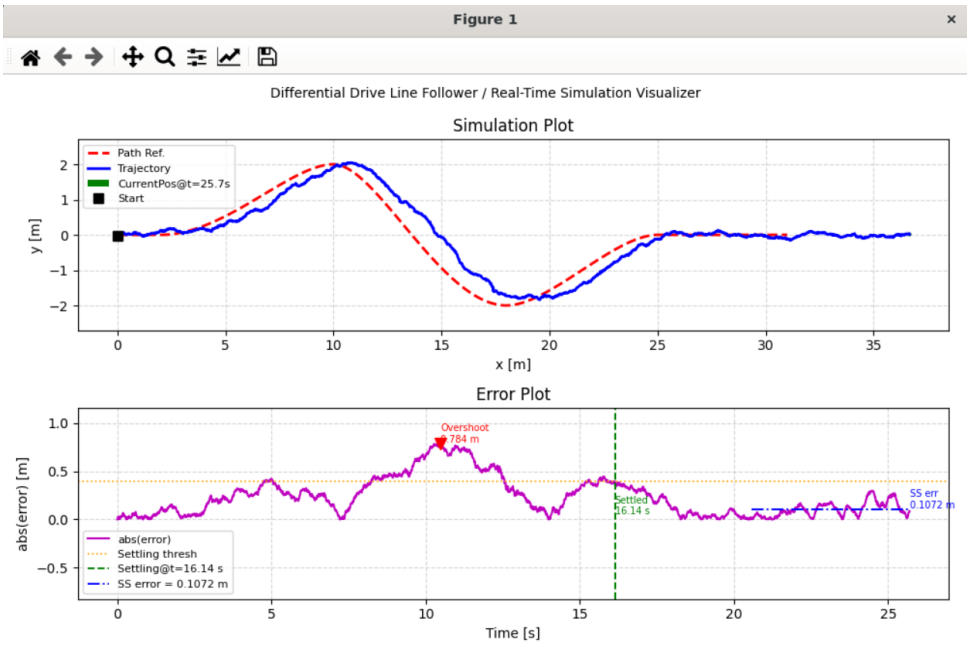

Figure 19: Curved path tracking with PID // Lateral Error Plots (Low-level noise applied)

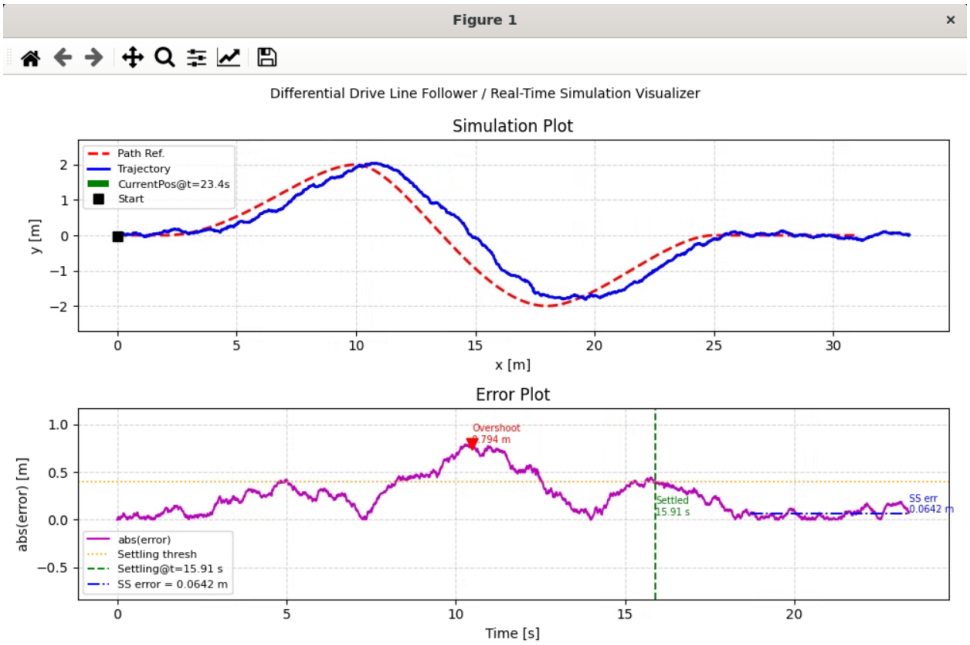

Figure 20: Curved path tracking with PD only [KI=0] // Lateral Error Plots (Low-level noise applied)

Metric	PD	PID
Overshoot (m)	0.794	0.784
Settling Time	15.91	16.14
Steady-State Error	0.0642	0.1072

Table 7: PD vs PID KPIs Comparison

Integral term effect: The addition of the integral term (PID) did not significantly change the main KPIs compared to the PD controller under the tested curved-path and noise conditions. Overshoot and settling time remained within a similar range, indicating that the proportional–derivative action already provided sufficient transient shaping.
However, the PD controller exhibited slightly higher sensitivity to noise, visible as increased oscillatory behavior in the lateral error signal. The integral term in the PID controller helped smooth small residual bias and reduced low-frequency steady-state fluctuations under noisy conditions.
Overall, while the integral action did not drastically improve the numerical KPIs, it improved robustness and reduced sensitivity to measurement disturbances.

Appendix – Source Code Repository

The complete source code, VSI configuration files, and experiment scripts for this project are available in the following public GitHub repository:

`https://github.com/Mostafa-Samy-Mohammed/LineFollowingRobotPID_DT.git`

The repository contains:

- lineFollowerTwin/ directory, which contains auto-generated Make files and src/ directory.
- VSI gateway configuration file (vsiBuildCommands).